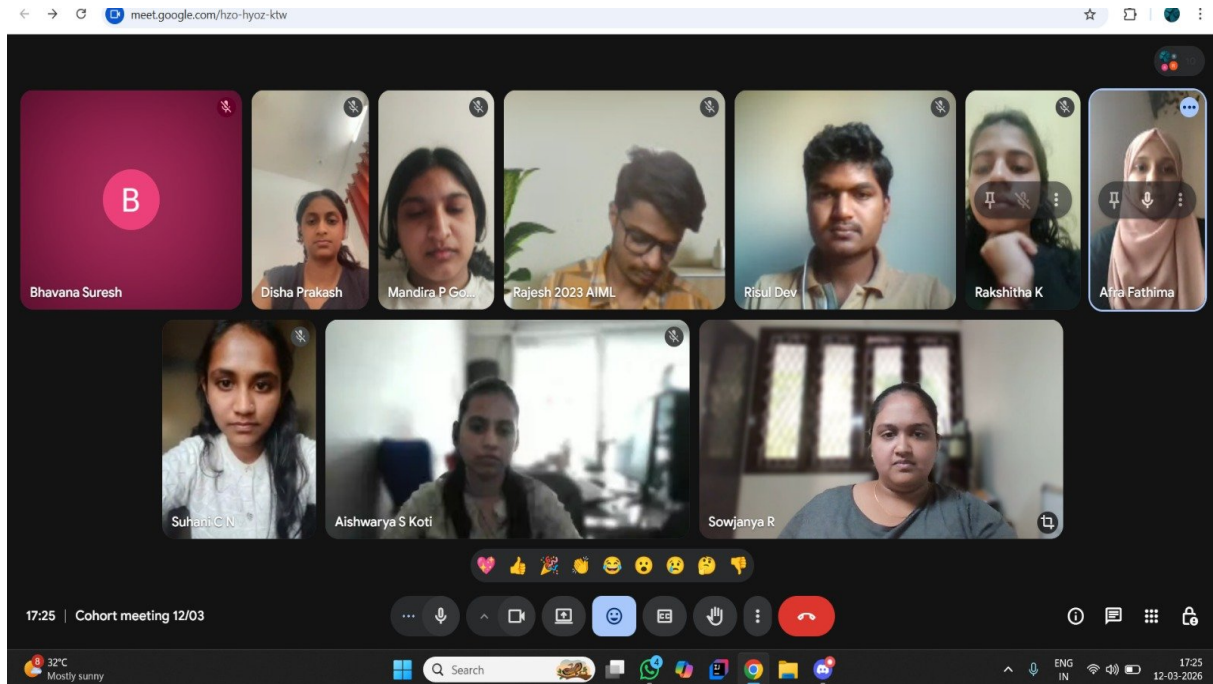


Mini Project: Cohort 21: Forest Fire Ignition Risk

Date: 12/03/2026

On Day 6 of the project, the team continued with the **model comparison and evaluation phase**. The **Evaluation Team** performed a detailed comparison between the **Decision Tree (DT)** and **Random Forest (RF)** classifiers, analyzing their outputs side by side. The **Data Team** independently completed and pushed all their preprocessing and feature engineering work to the shared repository, ensuring the pipeline was up to date for all members.



The Evaluation Team focused on running both classifiers on the same prepared dataset and comparing their performance metrics. Key indicators analyzed included the **accuracy score**, **confusion matrix**, and **classification report** for each model.

The **Decision Tree** model showed moderate accuracy but demonstrated signs of overfitting on the training data. Its confusion matrix revealed a higher number of misclassifications across fire risk categories, and the classification report indicated lower precision and recall for minority classes, limiting its generalization ability on unseen data.

The **Random Forest** model, being an ensemble method, consistently outperformed the Decision Tree across all evaluation metrics. It showed higher overall accuracy, fewer misclassifications in the confusion matrix, and improved precision, recall, and F1-scores in the classification report. Additionally, the Random Forest provided **feature importance** rankings, revealing which meteorological variables — such as temperature, wind speed, and relative humidity — contribute most to predicting forest fire ignition risk. The ensemble approach also significantly reduced the risk of overfitting by averaging predictions across multiple trees trained on different data subsets.

In parallel, the **Data Team** finalized all data preprocessing steps including handling of missing values, encoding of categorical variables, and normalization of numerical features. They also completed feature engineering and pushed the cleaned dataset along with all preprocessing scripts to the shared repository. Train/test splits were cross-verified to ensure reproducibility of all model evaluation results.

Key Discussion Points

Model Comparison: The team compared evaluation metrics of both models side by side. The Random Forest model showed clearly superior performance across all fire risk categories in the Algerian Forest Fire Dataset.

Why Random Forest Outperforms Decision Tree: A single Decision Tree is sensitive to noise and prone to high variance, whereas Random Forest builds multiple trees on bootstrapped subsets and averages their predictions, reducing variance and improving stability. For a safety-critical application like fire risk prediction, the lower false-negative rate of Random Forest makes it the clearly preferred model.

Data Team Contributions: All preprocessing and feature engineering outputs were successfully pushed to the shared repository, allowing the full team to work with consistent, up-to-date data for the final model pipeline.

Next Steps

Based on the Day 6 evaluation, the **Random Forest model** has been confirmed as the final model for the project. The next steps involve performing any remaining hyperparameter tuning, compiling all evaluation metrics and feature importance results into the project report, and preparing the final presentation for submission.